

Take-Home Assignment: Clinical Code Mapping from Free-Form Queries

Objective

Build a prototype that maps free-form clinical queries to appropriate standardized clinical concepts and codes using public/reference vocabularies.

Please use publicly available resources only. Do not use any proprietary code systems.

We are intentionally leaving the implementation approach open-ended. We are more interested in your reasoning, trade-offs, evaluation design, and ability to avoid clinically incorrect mappings than in a specific technology choice.

Sample Queries

Please test your approach on the following examples:

1. Patients with HbA1c above 7%
 2. Patients with fasting glucose greater than 126 mg/dL
 3. Patients with LDL cholesterol below 100 mg/dL
 4. Patients with eGFR less than 60 mL/min/1.73m²
 5. Patients with systolic blood pressure above 140 mmHg
 6. Patients diagnosed with type 2 diabetes
 7. Patients with chronic kidney disease stage 3
 8. Patients with heart failure
 9. Patients with COPD
 10. Patients with rheumatoid arthritis
 11. Patients currently taking metformin
 12. Patients treated with insulin
 13. Patients on GLP-1 receptor agonists
 14. Patients prescribed atorvastatin
 15. Patients exposed to systemic corticosteroids
 16. Patients who had a colonoscopy
 17. Patients with prior chemotherapy
 18. Patients with ECOG performance status 0 or 1
 19. Patients with BMI above 30
 20. Patients with a positive pregnancy test
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Assignment Requirements

For each query, your system should produce a structured mapping output that includes:

1. The interpreted clinical meaning of the query.

2. The relevant clinical entity or entities.
3. Any value, unit, temporal, or status constraints.
4. Candidate codes considered.
5. Final selected codes.
6. Codes or candidates rejected, with a brief reason.
7. A confidence score or ranking for the selected candidates.

The output format is up to you, but it should be easy to review programmatically.

What We Want to Understand

In your write-up, please explain:

1. How you interpreted the free-form query.
2. How you handled synonyms and alternative clinical phrasing.
3. How you retrieved candidate codes.
4. How you ranked and filtered candidates.
5. How you avoided clinically incorrect mappings.
6. How you evaluated the quality of your mappings.
7. What limitations remain in your approach.

Please keep the prototype practical. We do not expect a production-grade system, but we do expect clear clinical reasoning and thoughtful engineering trade-offs.

Deliverables

Please submit:

1. A short technical write-up.
 2. A working prototype, script, notebook, or small application.
 3. Instructions to run the code.
 4. Structured results for the 20 sample queries.
 5. A small evaluation table.
 6. A brief section on limitations and production improvements.
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Evaluation Criteria

Your submission will be assessed on:

Area	What we are looking for
Query interpretation	Can the system understand the clinical intent behind the free-form query?
Code retrieval	Does the system retrieve relevant candidate concepts with good recall?

Area	What we are looking for
Final mapping quality	Are the final selected codes clinically appropriate?
Precision	Does the system avoid misleading or overly broad mappings?
Synonym handling	Does the system handle abbreviations, variants, and related terminology?
Explainability	Can a reviewer understand why codes were selected or rejected?
Evaluation design	Is there a sensible way to measure mapping quality?
Scalability	Could the approach reasonably extend to larger vocabularies and more query types?

Public Resources You May Use

You may use any public or reference terminology sources, including but not limited to:

1. **OMOP Athena Vocabulary Search**
<https://athena.ohdsi.org/>
2. **OHDSI OMOP Standardized Vocabularies**
<https://ohdsi.github.io/TheBookOfOhdsi/StandardizedVocabularies.html>
3. **UMLS Documentation**
<https://documentation.uts.nlm.nih.gov/>
4. **LOINC Search**
<https://loinc.org/search/>
5. **RxNorm API Documentation**
<https://lhncbc.nlm.nih.gov/RxNav/APIs/RxNormAPIs.html>
6. **SNOMED CT Browser**
<https://browser.ihtsdotools.org/>
7. **ICD-10-CM Browser, CDC**
<https://www.cdc.gov/nchs/icd/icd-10-cm/browser-tool.htm>

You may also use embeddings, lexical search, APIs, downloaded vocabularies, LLMs, or any hybrid approach. Please document what you used and why.

Final Discussion Question

Assume that a proprietary clinical code system is later added, where each code includes:

- Code ID

- Display name
- Domain or category
- Synonyms, if available
- Parent/child relationships, if available
- Mapping to public vocabularies, if available

Briefly explain how your system would adapt to support this additional vocabulary while maintaining both high recall and high precision.